

Manish Kumar

SOFTWARE ENGINEER · PRODUCT BUILDER

Hyderabad, India

☎ (+91) 995-515-4740 | ✉ manishk.ayachi@gmail.com | 📷 [manishkr](#) | 🌐 [manishkay](#)

Summary

Systems and software engineer specializing in Rust, C++, Python, and CUDA. Expert in building high-performance, GPU-accelerated, low-latency systems, including analytical engines and data platforms. Proven track record delivering SaaS and desktop products end to end and leading engineering teams to build scalable, reliable, and commercially successful solutions.

Skills

Programming Languages Rust, C++, Python, Go, TypeScript

Databases Amazon DynamoDB, Apache Cassandra, MongoDB, MySQL, PostgreSQL, Redis, Scylla

Software Tools Actix, Arrow-rs, AWS Lambda, Axum, CUDA, DataFusion, Docker, Django, GraphQL, gRPC, Tokio, Velox

Work Experience

OPEN SOURCE CONTRIBUTOR

- Active contributor to Apache Arrow Rust (arrow-rs) and Apache DataFusion, focusing on query execution and compute kernels.
- Building CUDA-based JIT-compiled analytics engine in Rust for GPU-accelerated expression evaluation ([tachyon](#)).
- Building SIMD-optimized K-Nearest Neighbors vector search in Rust ([nyas](#)).

NeoCraft Technologies Pvt. Ltd.

[Hyderabad, India](#)

HEAD OF ENGINEERING

Oct. 2023 – Present

- Led architecture and development of high-performance backend systems for data-intensive and ML-driven products.
- Designed and delivered a recommendation engine using linear algebra and optimization techniques for a U.S.-based client.
- Migrated production workloads from AWS EC2 to AWS Lambda, reducing infrastructure costs by approximately 50 percent.
- Provided technical leadership, system design reviews, and hands-on implementation across multiple client engagements.

DataPelago India Pvt. Ltd.

[Hyderabad, India](#)

CONSULTANT/MEMBER OF TECHNICAL STAFF

Apr. 2024 – Sep. 2025

- Contributed to the development of a GPU-accelerated database query execution engine for analytical workloads.
- Implemented CUDA-based kernels for numerical expression evaluation, string processing, and type casting operations.
- Delivered 4x faster expression evaluation than NVIDIA cuDF through GPU kernel and execution pipeline optimizations.
- Collaborated closely with query planner and execution teams to optimize end-to-end query latency.
- Provided technical leadership and hands-on mentorship to engineering team.

Turing Inc.

[Hyderabad, India \(Remote\)](#)

SENIOR SOFTWARE ENGINEER

Apr. 2020 – Sep. 2023

- Contributed to a large-scale supply chain management platform for a U.S.-based enterprise client.
- Designed backend services powering a CRM used by thousands of enterprise users.
- Improved system performance by approximately 50 percent through query optimization and backend refactoring.
- Worked across distributed teams, contributing to design discussions, code reviews, and production support.

Setuserv Info. Pvt. Ltd.

[Hyderabad, India](#)

PRINCIPAL SOFTWARE ENGINEER

Aug. 2018 – Mar. 2020

- Architected an NLP-based text analytics SaaS platform using a microservices architecture.
- Built and operated scalable data pipelines using Kafka, Cassandra, and ScyllaDB on AWS.
- Designed REST APIs for external consumers and gRPC-based internal service communication.
- Developed a multi-dimensional analytics dashboard supporting thousands of query combinations.
- Improved overall data processing throughput by 2x through architectural and algorithmic optimizations.
- Led and mentored a team of six engineers through design, implementation, and delivery.

Ebizon Net Info Pvt. Ltd.

[Noida, India](#)

CTO AND TECHNICAL ARCHITECT

Jul. 2015 – Jul. 2018

- Founded and scaled PushFlew, a customer lifecycle SaaS platform serving SMBs.
- Built AppifyCart, a SaaS solution for creating and managing mobile commerce applications.
- Designed the backend architecture for Mofluid, a Magento-based mobile commerce platform.
- Owned system architecture, hiring, mentoring, and delivery for multiple concurrent products.

Geometric Ltd.

TECHNICAL SPECIALIST

- Developed a real-time 3D cloth simulation engine using NVIDIA CUDA for physics-based modeling.
- Implemented a geometric anomaly detection library for large-scale 3D engineering models.
- Led a team of five engineers across GPU computing and graphics-intensive projects.

Pune, India

Apr. 2010 – Jul. 2014

CSC India Pvt. Ltd. (PayPal IDC)

SOFTWARE ENGINEER

- Developed near real-time systems for fraud risk mitigation in PayPal payment workflows.
- Built a real-time address normalization and validation engine to improve data consistency.

Chennai, India

Oct. 2007 – Mar. 2010

Geometric Ltd.

SOFTWARE ENGINEER

- Developed surface modeling libraries for automated Class-A surface generation.
- Implemented optimization modules using Quadratic Programming techniques.

Pune, India

Jan. 2005 – Sep. 2007

IIT Bombay

JUNIOR RESEARCH FELLOW

- Researched optimization algorithms for automated Class-A surface design.
- Implemented Quadratic Programming models for CAGD applications using MATLAB.

Mumbai, India

Aug. 2004 – Jan. 2005

Education

IIT Bombay(Indian Institute of Technology Bombay)

B.TECH. IN COMPUTER SCIENCE AND ENGINEERING

Mumbai, India

Jul. 2000 - Aug. 2004